

Course Title: **Numerical Analysis**

Course No.: Math. Ed. 447

Level: BICTE

Semester: Fourth

Nature of course: Theoretical

Credit Hour: 3 hours

Teaching Hour: 48 hours

1. Course Description:

This course is designed for the students of BICTE under Tribhuvan University. It helps students to fulfill their increasing desire towards numerical answers to applied problems with the help of methods and techniques of numerical analysis. Although numerical methods have always been useful, their role in the present day scientific research is of fundamental importance. It deals with numerical methods which give the solution when ordinary analytical methods fail for the solution of transcendental equations. In addition, it deals those numerical techniques which can be used for the solution of system of linear equations through matrix computations along with solution of non-linear equations through interpolation and iterative method of differentiation and integration. This course also provides a foundation for the mathematical modeling in the field of research.

2. General Objectives

- To understand errors and approximation.
- To use different methods for solving transcendental and linear simultaneous equations.
- To define different types of differences and construct their tables, and establish the relationship between them.
- To be familiar with interpolation and apply suitable interpolation formula for numerical problems
- To deal with numerical approximations of derivatives
- To approximate computation of an integral using numerical techniques

3. Specific Objectives and contents

Unit-I Errors and computation of roots (8 hrs)	
• To identify the types of errors • To derive general error formula • To generalize a series approximation • To solve linear equations graphically • To find solution of equations by bisection method • To discuss the method of false position • To solve equations by iteration method • To use Newton –Raphson iteration formula • To apply the Muller ‘s method to approximate the roots of equations	• Significant digits • Errors • General error formula • Error in a series approximation • Linear equations • Graphical solution of equations • Bisection method • The method of false position • Iteration method • Newton – Raphson method • Generalized Newton’s formula for multiple roots • Muller’s method
Unit –II Solution of Linear simultaneous Equations (8 hrs)	
• To identify linear-simultaneous equations. • To apply Gauss elimination method in solving simultaneous equations • To solve simultaneous equations by Gauss –Jordan method • To solve LS equations by using Jacobi’s and Gauss – Seidel iteration method • To discuss and use factorization, Iterative and partition methods to solve simultaneous equations.	• Linear simultaneous equations (LSE) • Gauss elimination method • Gauss – Jordan method • Jacobi – iteration method • Gauss – Seidel iteration method • Matrix inversion method • Factorization method • Iteration method • Partition method

Unit –III Differences of polynomials (10 hrs)	
• To discuss forward and backward difference operators • To construct difference tables • To discuss properties of the forward difference operator • To establish relationship among the operators E, D and Δ • To express a given polynomial in factorial notation • To identify the central difference operator and the mean operator • To construct the central difference table • To establish relationship between the operators Δ, ∇, E, μ and δ	• Forward difference operator • Forward difference table • The operator E • Relation between the operators E and Δ • The operator D • Backward difference table • Factorial polynomial • Central difference operator • Central difference table • Mean operator • Relationship between operators Δ, ∇, E, μ and δ
Unit –IV Interpolation with Equal Intervals (8 hrs)	
• To derive and use Newton – Gregory forward interpolation formula • To derive and use Newton –Gregory backward interpolation formula • To apply forward and backward interpolation formulae in solving problems • To derive and use Gauss’ forward and backward interpolation formula • To apply Bessel’s and Stirling's formula for interpolation	• Newton –Gregory forward interpolation formula • Newton - Gregory backward interpolation formula • Error in the interpolation formula • Gauss’ forward interpolation formula • Gauss’ s backward interpolation formula • Bessel’s formula • Stirling’s formula
Unit -v Interpolation with Unequal Intervals (4 hrs)	
• To discuss linear and quadratic interpolations • To find divided differences • To establish the relationship between divided differences and ordinary differences	• Linear interpolation • Quadratic interpolation • Divided differences • Second divided difference • Newton ’s divided difference interpolation • Relation between divided differences and ordinary differences
Unit – VI Numerical Differentiation and integration (10 hrs)	

• To derive formula for the derivative using forward and backward difference formula • To derive formula for derivative using central difference formula • To derive general quadrature formula • To apply trapezoidal rule , Simpson's one –third rule ,three-eighth rule , Bool's rule and Weddle's rule for solving numerical problems • To find errors in quadrature formula	• Numerical differentiation • Derivative using forward difference formula • Derivative using backward difference formula • Derivative using central difference formula • General quadrature formula for equidistant ordinates • Trapezoidal rule • Simpson,s One –Third rule • Simpson,s Three – Eighth rule • Bool,s rule • Weddle ,s rule • Errors in quadrature formula
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4.Instructional Techniques

Units	Activity and Instructional Techniques
Unit I	• Individual and group discussion on calculating errors
Unit II	• Individual and group discussion on bisection and iteration methods • Group and individual assignments on problems of getting roots by bisection method
Unit III	• Group and individual discussion on different methods of solving linear simultaneous equations
Unit IV	• Individual and group assignments on finite differences
Unit V	• Presentation and discussion on computer programming in c++ of important methods
Unit VI	• Individual and group presentation on divided differences and ordinary differences

5. Evaluation

Internal evaluation

Internal evaluation will be conducted by course teacher based on following activities:

a. Attendance	5 points
b. Participation in learning activity	5 points
c. First assessment test	10 points
d. Second assessment test	10 points
e. Third assessment test	10 points

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Total 40 points

NOTE: Internal evaluation and assignments may include the numerical calculation and computation by using different computer application like as Matlab, Geogebra and MS Excel also.

External Evaluation:

Faculty of Education, Examination division will conduct final examination of weight 60 points at the end of semester. This 60 points is divided in final examination paper as

Objective questions (10 x 1)	10 points
Short answer questions (6 x 5)	30 points
Long answer questions (2x 10)	20 points

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Total 60 points

6.Recommended and Reference Books

6.1 Recommended books

Sastry, S.S. (1990). *Introductory methods of numerical analysis*, New Delhi :Prentice-Hall of India (Units I – VI)

Gupta S. and Sharma S.(2014).*Numerical analysis*, New Delhi : S.K .Kataria &Sons (Units I – VI)

6.2 Reference books

Conte S.D. (1965) , *Elementary numerical analysis* Mc Graw- Hill Froberge

C.E. (1965) , *Introduction to numerical analysis* ,Adison Wesley

Jian , M.K.(1971) , *Numerical analysis for scientists and engineers* Delhi:S.B.W . Publishers

Sastry S.S. (1997) , *Engineering mathematics* , New Delhi : Prentice-Hall of India

Stanton , R.G. (1967) , *Numerical methods for science and engineering* , New Delhi :Prentice-Hall of India